

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Hard

Question

For a particular car, the linear function f gives the predicted power, in brake horsepower (**bhp**), for engine speeds between **1,000 revolutions per minute (rpm)** and **6,000 rpm**. According to this function, the car's predicted power is **433 bhp** at an engine speed of **3,331 rpm** and **600 bhp** at an engine speed of **4,500 rpm**. The equation $f(x) = \frac{1}{7}(x - a) + 433$ defines f , where x is the engine speed, in **rpm**, and a is a constant. What is the value of a ?

Correct Answer: 3331

Rationale

The correct answer is **3,331**. For engine speeds between **1,000 revolutions per minute (rpm)** and **6,000 rpm**, it's given that the equation $f(x) = \frac{1}{7}(x - a) + 433$ defines the linear function f and gives the predicted power, in brake horsepower (**bhp**), at an engine speed of x **rpm**, where a is a constant. It's also given that the car's predicted power is **433 bhp** at an engine speed of **3,331 rpm**. Substituting **3,331** for x and **433** for $f(x)$ in the equation yields $433 = \frac{1}{7}(3,331 - a) + 433$. Subtracting **433** from both sides of this equation yields $0 = \frac{1}{7}(3,331 - a)$. Multiplying both sides of this equation by **7** yields $0 = 3,331 - a$. Adding a to both sides of this equation yields $a = 3,331$. Thus, the value of a is **3,331**.